Smart Power-Failure Predictor ASIC

50% Project Progress Report — Phase 1: Microarchitecture, RTL Design, Simulation & Logic Synthesis

PROJECT MILESTONE STATUS: 50% COMPLETED (PHASE 1 / MID-TERM REVIEW) **STAGE: FRONT-END COMPLETE**

Target Technology:	SkyWater 130nm CMOS (`sky130_fd_sc_hd`)	**Review Milestone:**	50% Progress Evaluation (Phase 1)
Design Flow:	Pure Software EDA Flow (Zero Hardware)	**RTL Status:**	100% Complete & Modular (Verilog-2005)
Simulation Verification:	8 / 8 Tests Passed (100% Clean)	**Logic Synthesis:**	Mapped 1,296 Standard Cells (Yosys)
Phase 1 Deliverables:	RTL Modules, Testbench, VCD Waveforms, Netlist	**Phase 2 Scope:**	Floorplan, CTS, PnR Routing, DRC/LVS Tapeout

1. Project Execution Roadmap & 50% Milestone Tracker

The project is structured into two equal 50% phases. **Phase 1 (Front-End Design & Verification)** is 100% completed and documented below. **Phase 2 (Physical Implementation & Signoff)** represents the planned roadmap for the final review.

Stage	ASIC Design Phase	Tools Used	Target Deliverable	Status
1	Architecture & Math Formulation	Analytical Model	Sampling & dV/dt Derivative Equations	COMPLETED (100%)
2	Microarchitecture & Register Map	Specification	Memory-Mapped CSR Bank (0x00 to 0x06)	COMPLETED (100%)
3	Synthesizable Verilog RTL Design	Verilog HDL	Top-level + 4 Sub-blocks (Modular RTL)	COMPLETED (100%)
4	Functional Simulation Verification	Icarus Verilog + GTKWave	8/8 Scenarios Passed, VCD Waveforms	COMPLETED (100%)
5	Logic Synthesis & Gate Mapping	Yosys + ABC Engine	Gate Netlist (1,296 Cells, 159 DFFs)	COMPLETED (100%)
6	Floorplanning & Power Grid (PDN)	OpenROAD / PnR Flow	Die Size (240x240 um), Met4/Met5 Rings	PLANNED (Phase 2)
7	Standard Cell Placement	OpenROAD Placer	Global & Detailed Cell Placement	PLANNED (Phase 2)
8	Clock Tree Synthesis (CTS)	TritonCTS	Balanced H-Tree, Skew < 100 ps	PLANNED (Phase 2)
9	Detailed Routing & Parasitics	TritonRoute	Multi-layer Metal Routing & SPEF	PLANNED (Phase 2)
10	DRC/LVS Signoff & GDSII Mask	Magic / KLayout	Silicon Mask Signoff & 3D Layout	PLANNED (Phase 2)

2. Motivation & Architectural Innovation

The Flaw in Traditional Brownout Detectors (BOD): Standard BOD modules in microcontrollers only assert an interrupt when voltage drops below the absolute minimum operational limit ($V < V_{\text{BOD}} \approx 2.5\,\text{V}$). By that moment, the CPU oscillator has already begun to jitter, internal registers scramble, and the processor has **zero microseconds** to write critical sensor logs or state variables into non-volatile Flash/EEPROM.

The Smart Power-Failure Predictor Innovation: Our ASIC observes continuous 12-bit streaming voltage samples ($V[t]$) and performs predictive dual-mechanism anomaly detection:

• **Negative Rate-of-Fall (dV/dt) Engine:** Calculates $\Delta V / \Delta t = V[t-1] - V[t]$. If slope exceeds SLOPE_THRESH ($50\,\text{mV/sample}$), a catastrophic disconnect (loose battery, severed trace, capacitor collapse) is caught instantly.

• **Moving Trend Filter:** A 4-tap FIR averaging pipeline $\frac{1}{4}\sum_{k=0}^3 V[t-k]$ eliminates high-frequency inductive switching spikes while tracking genuine battery decay.

• **Two-Tier Early Warning:** Asserts pfi_warn_o at $2.90\,\text{V}$ (Tier 1 Warning), giving the MCU **10 to 50 milliseconds of advance notice** to save state before voltage reaches the panic threshold $2.60\,\text{V}$ (Tier 2 Critical).

3. Front-End Verilog RTL Implementation

The Verilog RTL is structured into 4 fully synchronous modular sub-blocks coordinated by a central top-level chip wrapper (rtl/spf_predictor_top.v):

Register	Address	Type	Reset Value	Functional Description
CTRL_REG	0x00	R/W	0x000F	Bit 0: Enable, Bit 1: Slope En, Bit 2: Filter En, Bit 3: Auto-Recover
V_WARN_THRESH	0x01	R/W	2900 mV	Warning trip voltage threshold (triggers pfi_warn_o interrupt)
V_CRIT_THRESH	0x02	R/W	2600 mV	Critical panic voltage threshold (triggers shdn_req_o shutdown)
SLOPE_THRESH	0x03	R/W	50 mV/smp	Max negative dV/dt threshold (detects sudden disconnects instantly)
HYST_THRESH	0x04	R/W	100 mV	Recovery hysteresis window (prevents state chattering during recovery)

STATUS_REG	0x05	RO	0x0000	Live status: [0]=Warn, [1]=Shutdown, [4:2]=FSM State, [5]=Slope Trip
INT_CLEAR	0x06	WO	0x0000	Write-1 to clear sticky warning, shutdown, and slope interrupt flags

4. Phase 1 Verification Results: Simulation & Testbench

A self-checking Verilog verification testbench (tb/spf_predictor_tb.v) was simulated using **Icarus Verilog (`iverilog`)**. Waveforms were recorded to sim/power_sag_dump.vcd and analyzed in GTKWave. All 8 verification test scenarios passed with zero errors:

Test Scenario	Simulated Real-World Condition	Observed Hardware Reaction	Verification Result
Test 1A: Nominal Power	Steady 3.30V nominal power rail with baseline sensor noise	State remains NORMAL (Green LED, pfi_warn_o = 0, shdn_req_o = 0)	**PASSED (Clean)**
Test 1B: Inductive Ripple	Single-cycle switching noise dip (30 mV ripple, 3270 mV)	4-tap moving filter absorbs ripple; zero false alarms triggered	**PASSED (Noise Immune)**
Test 2: Gradual Battery Sag	Slow battery drain (decay rate 35 mV/smp < 50 mV slope limit)	Early Warning fires at 2845 mV! Advance notice delivered while V > 2.6V	**PASSED (Early Alert)**
Test 3: Critical Power Drop	Severe collapse crossing critical threshold (2550 mV < 2600 mV)	shdn_req_o asserts immediately; Red LED indicates emergency shutdown	**PASSED (Shutdown Safe)**
Test 4: Recovery Hysteresis	Power restored from 2.55V back to 3.30V across hysteresis bound	State transitions smoothly through RECOVERY to NORMAL without chatter	**PASSED (Hysteresis OK)**
Test 5: Catastrophic dV/dt	Sudden capacitor disconnect (instant 350 mV drop in 1 sample)	Derivative engine catches slope instantly; fires shutdown BEFORE 2.90V!	**PASSED (Fast Slope Trip)**
Test 6A: CSR Write & Read	Dynamically configure V_WARN = 3100 mV, SLOPE = 100 mV over bus	Registers successfully updated and verified through readback bus cycle	**PASSED (Config OK)**
Test 6B: Interrupt Clear	Write 0x0007 to INT_CLEAR register to clear latched flags	Sticky interrupt flags cleared; status register confirms clean return	**PASSED (Clear Confirmed)**

5. Preliminary Gate-Level Logic Synthesis (Yosys)

Using the **Yosys Open SYnthesis Suite** with the ABC technology mapping engine, the synthesizable Verilog modules were mapped into standard logic gates. The preliminary gate netlist (syn/spf_synth_netlist.v) contains **1,296 standard cells** with zero dangling nets or connectivity errors:

Cell Class	Component Name	Cell Count	Percentage	Function in Microarchitecture
Sequential	$_DFF_PN0_ / PN1_	159	12.3 %	FSM state registers, CSR storage, filter delay taps
Combinational	$_NAND_ / $_AND_	688	53.1 %	Adder carry chains, comparator logic, bus decoders
Multiplexers	$_MUX_	141	10.9 %	Bus readback muxing & FSM next-state branches
Logic / Arithmetic	$_XOR_ / $_XNOR_	110	8.5 %	12-bit arithmetic subtractor & derivative engine
Conditionals	$_OR_ / $_NOR_	157	12.1 %	Interrupt flag ORing and threshold comparator flags
Inverters	$_NOT_	41	3.1 %	Signal inversion and reset distribution buffers
TOTALS	**Sky130 Mapped Cells**	**1,296**	**100.0 %**	**Clean Gate Netlist, 0 Synthesis Violations**

6. Phase 2 Scope & Work Planned for Final Review (Remaining 50%)

With the front-end design, RTL, testbench simulation, and synthesis successfully locked in Phase 1, the remaining 50% of the project focuses on **back-end physical design, routing, and tapeout signoff**:

- **Milestone 6: Physical Floorplanning & Power Mesh (PDN):** Define core area ($180 \times 180\,\mu\text{m}$) within die boundary ($240 \times 240\,\mu\text{m}$), create dual-perimeter $V_{\text{DD}}/V_{\text{SS}}$ power rings, and route vertical straps (Met4).
- **Milestone 7: Standard Cell Placement:** Global and detailed placement of all 1,296 standard cells into 60 cell rows with row flipping for power rail sharing.
- **Milestone 8: Clock Tree Synthesis (CTS):** Synthesize a balanced H-Tree clock network on Met3/Met4 targeting $< 100\,\text{ps}$ clock skew across all 159 D-Flip-Flops.
- **Milestone 9: Detailed Routing & Parasitic Extraction:** Multi-layer metal routing (Met1 through Met5) and RC parasitic SPEF extraction for post-layout STA.
- **Milestone 10: Physical DRC/LVS Verification & Final Tapeout:** Verify zero design rule violations, confirm 100% netlist matching, and generate the final silicon GDSII mask file for foundry fabrication.

PHASE 1 (50%) MILESTONE EVALUATION SUMMARY:
All Phase 1 goals have been achieved with 100% success. The RTL design is synthesizable, mathematically verified across 8 real-world power sag and noise scenarios, mapped to 1,296 standard cells in Yosys, and completely documented. The project is fully on schedule to complete Phase 2 physical design and final tapeout signoff.